

Isaac Essential Physics
Step into Physics
Draft Waves Excerpt

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Light Sources and Rays

Light enables us to **see** things.

Light is made by **light sources**.

We see a light source if its light ends up in **our eyes**.

- 1 Is it a light source? How did you decide?
- | | |
|----------------------------|-------------------------|
| (a) A TV screen, | (d) the Sun, |
| (b) a desk, | (e) a sheet of paper, |
| (c) a mobile phone screen, | (f) a poster on a wall. |

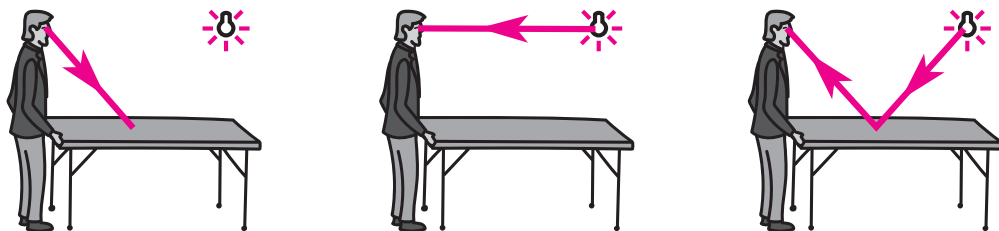
A cup is **not** a light source. We can see it when

- light from a light source
- travels to the cup,
- then **bounces off** the cup.
- The light then travels from the cup to **our eyes**.

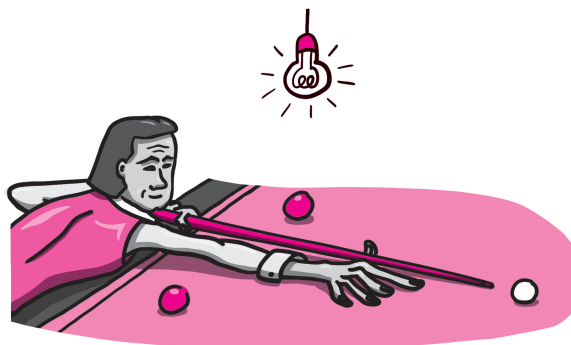
When light bounces off something, we say it **reflects**.

Light travels through the air in a **straight line**.

- 2 A person can see a table in a room with a light bulb. Which of the following diagrams is correct? The lines with arrows show the light travelling.



- 3 Draw lines with arrows to show how a snooker player is able to see a ball.



The straight lines with arrows on our diagrams are called **rays**.

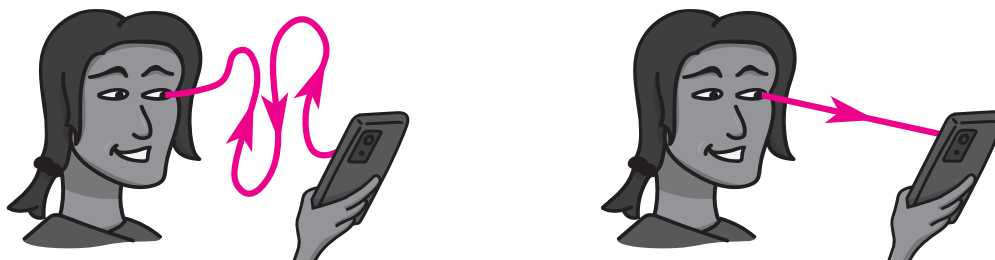
2

- 4 A lifeguard on the beach sees a swimmer in need of help. Complete the sentences using the words **reflects**, **lifeguard**, **swimmer**, **Sun**, **eyes**. You can use a word more than once.

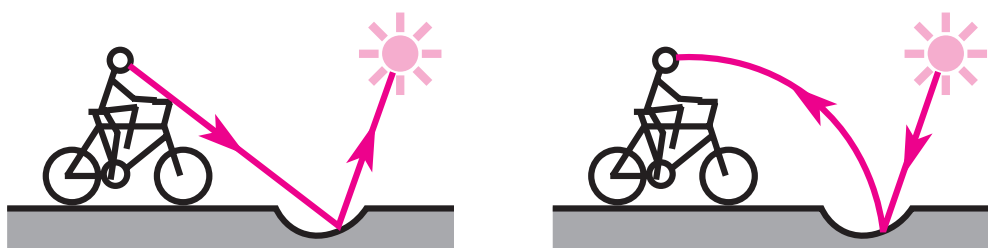
Light from the _____ travels to the _____ and _____ off them.

Light then travels from the _____ to the _____ and ends up in their _____.

- 5 What is wrong with these diagrams showing a person looking at their phone?



- 6 What is wrong with these diagrams showing a cyclist looking at a hole in the road?

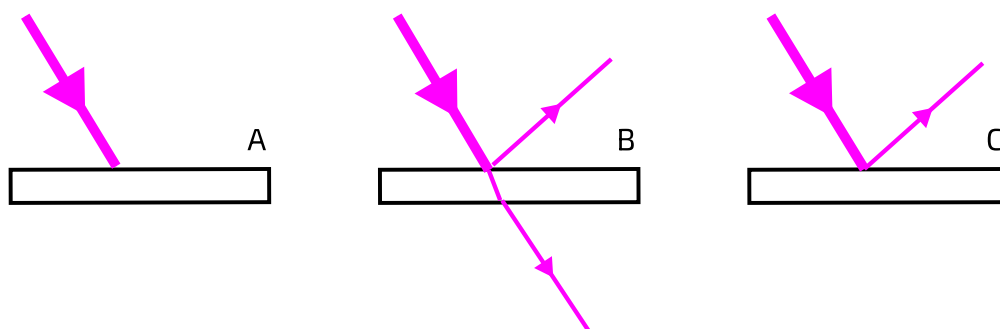


Light can not pass through an **opaque** material.

When light reaches an opaque object, there is **reflection** and **absorption**.

The **energy** from the light that is absorbed causes the material to warm up.
This increases the **thermal** energy store of the material.

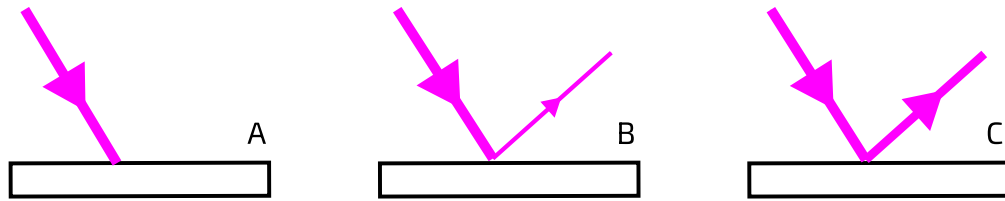
- 7 Which of the diagrams below show light reaching an opaque material?



8 In these diagrams, the thickness of the line shows the brightness of the light.

(a) Which of these diagrams show reflection?

(b) Which of these diagrams show absorption?



9 Complete the sentences using the words **reflected, absorbed, some, no, warmer, cooler, slowly, quickly, bright**. You can use a word more than once. Some words are not used

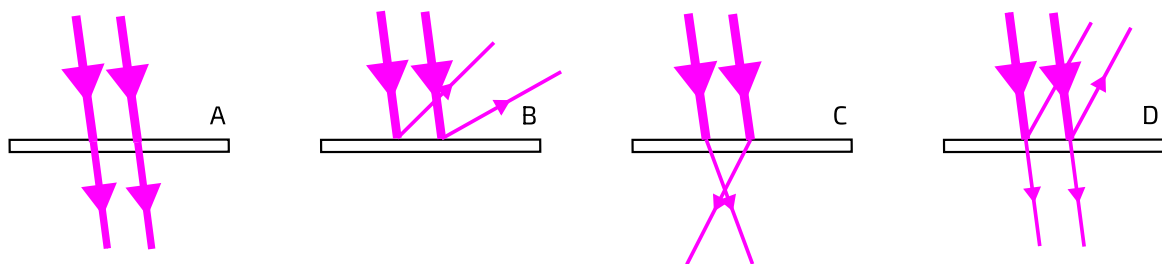
My desk is opaque and painted a pale colour. When sunlight shines on it, _____ light goes through it. I can see the desk, because most of the light is _____ by the surface. After the desk has been in sunlight for an hour, it feels much _____. This means that some sunlight was _____.

I paint my desk a darker colour. Less light will now be _____ off it, and so it won't look as _____. The desk will now warm up more _____ in the sunlight because more light is now being _____.

Light can go through a thin white sheet. Light can also go through a glass window. You can see what is on the other side of a glass window. You can't see through the thin white sheet clearly because the rays of light get **scattered (diffuse transmission)**.

The glass window is **transparent**. The thin white sheet is **translucent**.

10 Match the diagrams to the words **opaque, translucent** and **transparent**.



11 Do we need an opaque, transparent or translucent material? Fill in the table.

Description	Opaque, transparent or translucent?
The lens in a camera	
The window in a bathroom	
Bedroom curtain if you need to sleep during the day	

Light Sources and Rays Practice

If an object does not emit light, we can only see it because light bounces off it.

- 1 Light sources produce their own light. If an object does not emit light, we can only see it because light bounces off it.
Decide whether each object is a light source or not.

(a) a fire

(d) the Moon

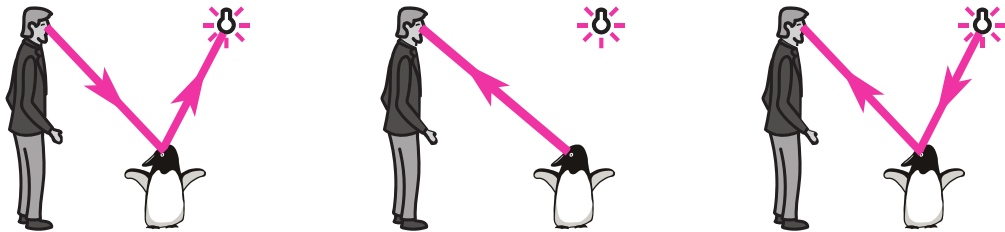
(b) a mirror

(e) a lit match

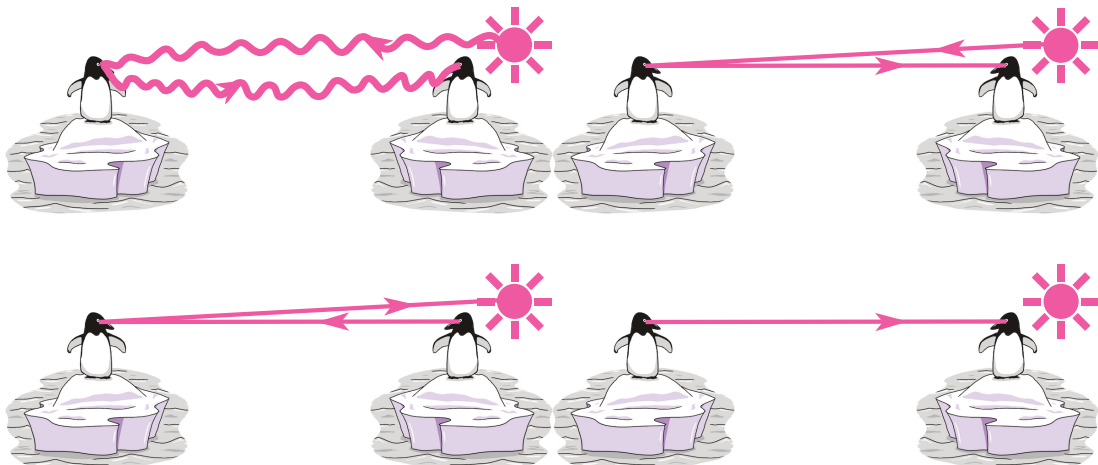
(c) a computer screen

(f) a shiny spoon

- 2 A person can see a penguin in a room with a light bulb. Which of the diagrams below is correct? Circle your answer.



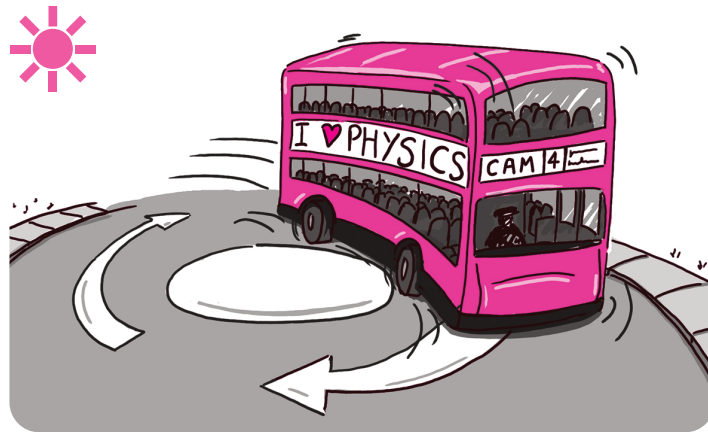
- 3 In the diagrams below, it is sunset and the penguin on the right is looking at its penguin friend on the left. Which of the following diagrams correctly represents how they see the other penguin?



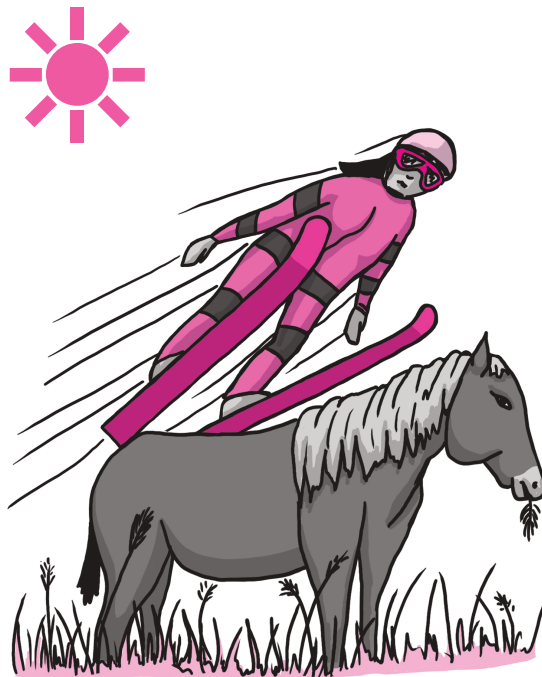
- 4 A student goes to a concert and sees their favourite musician perform! Complete the sentences using the words **reflects**, **student**, **musician**, **spotlight**, **eyes**. You can use a word more than once.

Light from the _____ travels to the _____ and _____ off them. Light then travels from the _____ to the _____ and ends up in their _____.

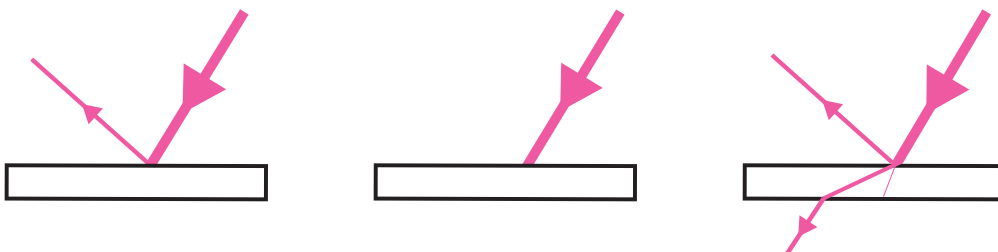
- 5 Draw rays to show how the bus driver sees the roundabout.



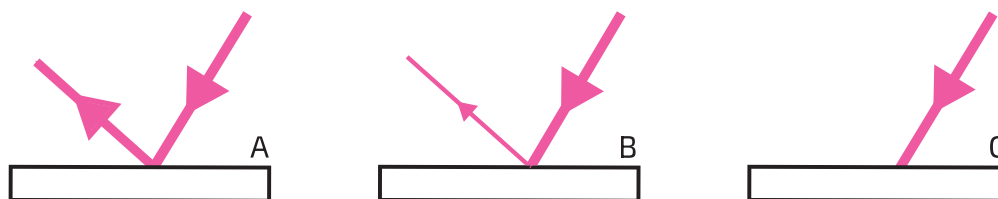
- 6 Draw rays to show how the skier sees the horse's ear.



- 7 Which of the diagrams below correctly shows light reaching an opaque material?



- 8 In these diagrams, the thickness of the line shows the brightness of the light.



(a) Which of these diagrams correctly shows reflection?

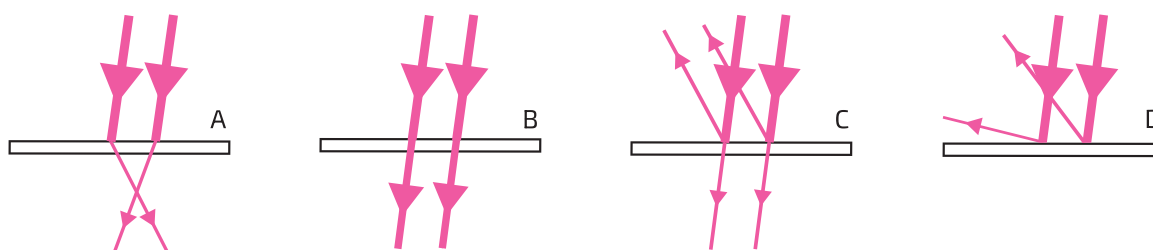
(b) Which of these diagrams correctly shows absorption?

- 9 Complete the sentences using the words **reflected**, **absorbed**, **some**, **no**, **warmer**, **cooler**, **slowly**, **quickly**, **bright**. You can use a word more than once. Some words are not used.

(a) My bike is opaque and painted pale blue. When sunlight shines on it, _____ light goes through it. I can see the bike, because most of the light is _____ by the surface. After the bike has been in sunlight for an hour, it feels much _____. This means that some sunlight was _____.

(b) I paint my bike dark blue. Less light will be _____ off it, and so it won't look as _____. The bike will now warm up more _____ in the sunlight because more light is now being _____.

- 10 Label each diagram as either opaque, translucent or transparent



- 11 Do we need an opaque, transparent or translucent material? Fill in the table.

Description	Opaque, transparent or translucent?
A sleep mask	
A paper globe lampshade for an overly bright lightbulb	
Lenses in reading glasses	

Reflection

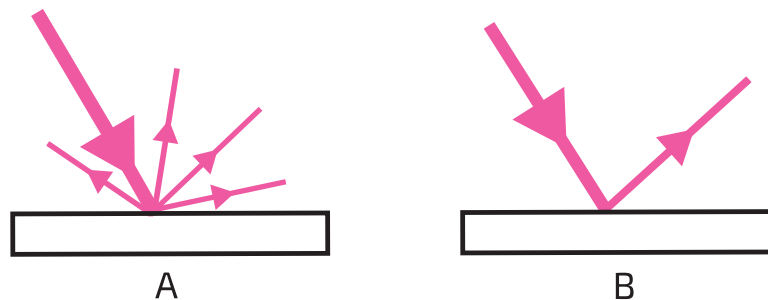
When light bounces off something, we say it **reflects**.

Light travels through the air in a **straight line**.

Depending on the **surface** light can reflect in two ways.

- in **specular** reflection, light only leaves the surface in **one direction**.
- in **diffuse** reflection, light leaves the surface in **all directions**.

1 The diagrams below show the two ways light can reflect.



(a) Which diagram shows **specular reflection**?

(b) Which diagram shows **diffuse reflection**?

2 Which kind of reflection is described in each case? Fill in the table.

Description	Specular or Diffuse?
Reflection in a mirror	
Light bouncing off a painted wall	
Reflection in a glass window	
Light bouncing off a friend's clothes	

3 If you want to see an image of yourself, which kind of reflection do you need?

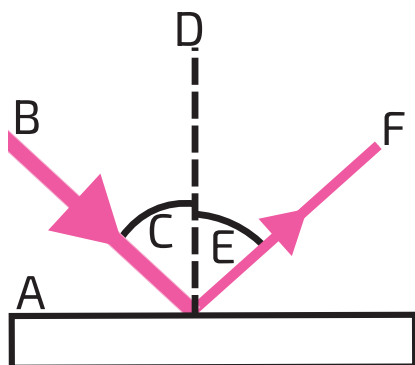
4 The brightness of reflected rays depend on the type of reflection. Complete the sentences using the words **weaker**, **energy**, **direction**, **specular**, **diffuse**.

Light from a _____ reflection can be almost as bright as the incoming ray. This is because nearly all of the _____ of the ray is going the same way.

Light going in one direction from a _____ reflection will be much _____ than the incoming ray, because only a small fraction of the energy travels in that _____.

A line drawn at **right** angles to the surface is called the **normal**.

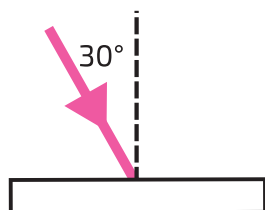
5 Which words describe the features in the diagram below? Fill in the table.



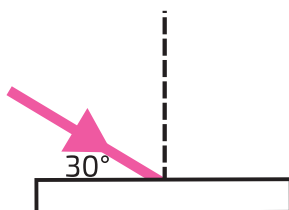
Technical term	A, B, C, D, E or F?
surface	
reflected ray	
incident ray	
angle of reflection	
angle of incidence	
normal	

6 In each diagram below, write down the angle of incidence.

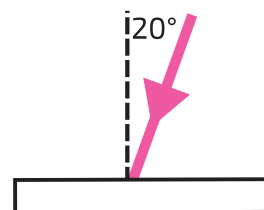
(a)



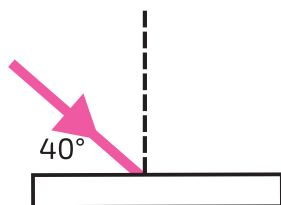
(b)



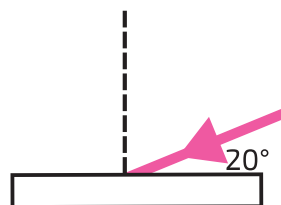
(c)



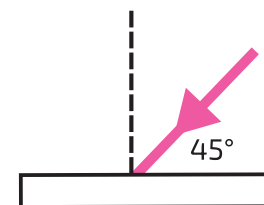
(d)



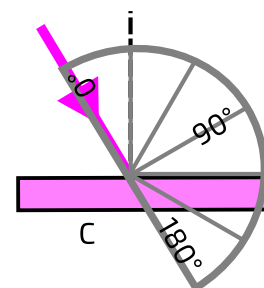
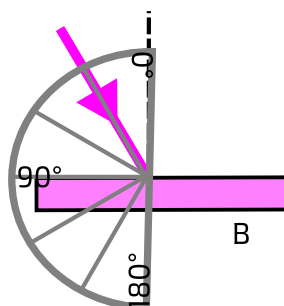
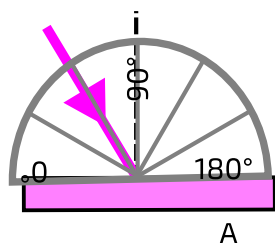
(e)



(f)



7 The diagrams below show three attempts to measure an angle of incidence.



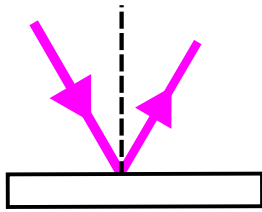
(a) From which of these protractors could we read off the correct angle?

(b) State the angle of incidence of the ray.

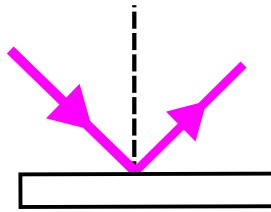
(c) What do we line the centre of the protractor up with?

- 8 Measure the angles of incidence and the angles of reflection using a protractor.

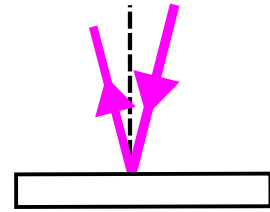
(a)



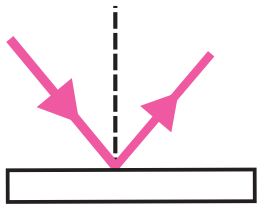
(b)



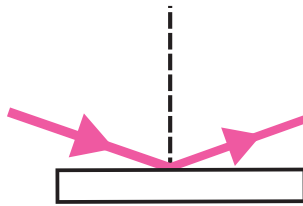
(c)



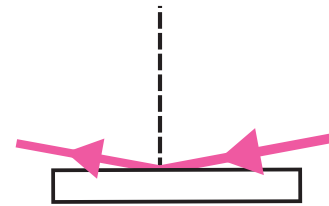
(d)



(e)

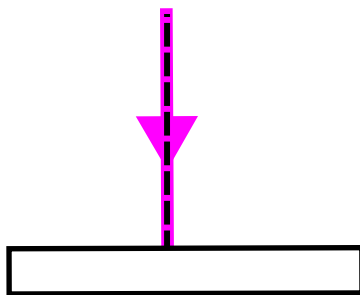


(f)



- 9 Look at your answers to question 8. How do you predict the angle of reflection if you are told the angle of incidence?

- 10 Look at the diagram below.

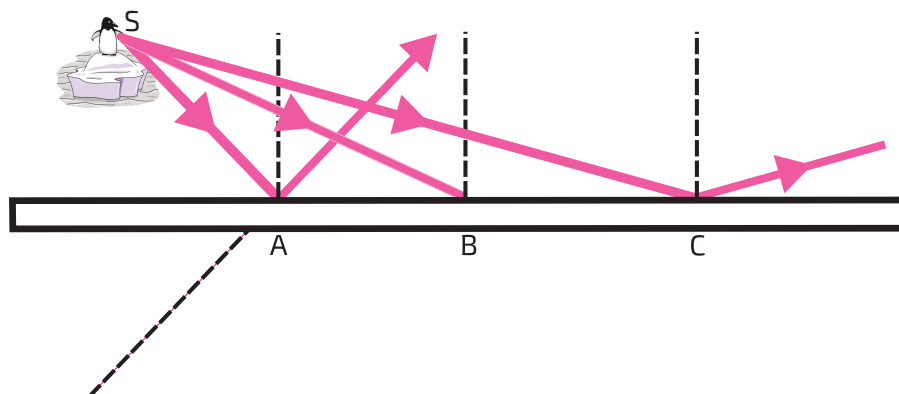


(a) State the angle of incidence.

(b) State what the angle of reflection will be.

(c) Draw the reflected ray on the diagram.

- 11 The diagram below shows three rays from the same point striking a mirror.



(a) Measure the angle of incidence of the ray going to B.

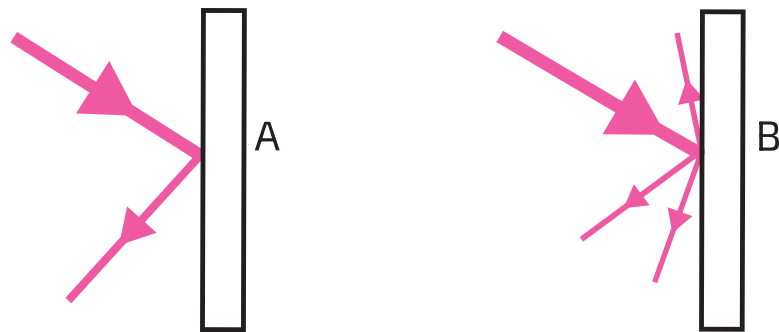
(b) Draw the reflected ray from B on the diagram using a protractor and ruler.

(c) Continue the reflected rays from B and C through the mirror with dashed lines. Use a ruler. The line from A has been done for you as an example.

(d) Where do the dashed lines meet? Describe the location in words.

Reflection Practice

- 1 The diagrams below show the two ways light can reflect.



(a) Which diagram shows specular reflection?

(b) Which diagram shows diffuse reflection?

- 2 Which kind of reflection is described in each case? Fill in the table

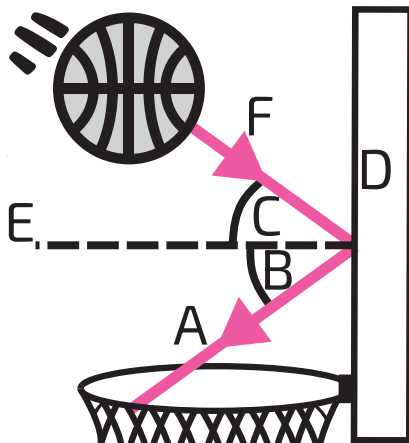
Description	Specular or Diffuse?
Seeing your face in a polished metal spoon	
Light bouncing off a sheet of printer paper	
Light bouncing off a dry concrete pavement	
Sunlight reflecting off a perfectly still puddle of water	

- 3 Your housemate always takes ages in the bathroom and you suspect they're spending a long time staring at themselves in the mirror. You 'clean' the mirror with a dirty cloth and now reflections are blurred. What kind of reflection is this?
- 4 The brightness of reflected rays depend on the type of reflection. Complete the sentences using the words **rough**, **diffuse**, **scattered**, **smooth**, **specular** and **direction**.

In a large lecture theatre, the back walls are often fitted with _____, jagged acoustic panels to encourage _____ reflection. This ensures that sound is _____ randomly around the room, preventing a distracting, bouncing echo.

On the other hand, in a mosque, the _____, curved section of the wall called the mihrab is designed for _____ reflection. This acts like a mirror for sound, focusing the speaker's voice in a specific _____ toward the worshippers so that the message remains loud and clear.

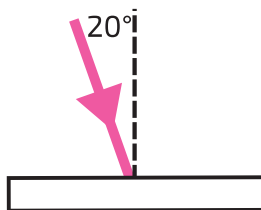
- 5 The diagram below shows a basketball being shot off a backboard. Which words describe the features of the diagram? Fill in the table.



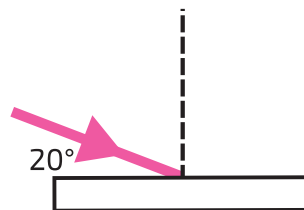
Technical term	A, B, C, D, E or F?
surface	
reflected ray	
incident ray	
angle of reflection	
angle of incidence	
normal	

- 6 In each diagram below, write down the angle of incidence.

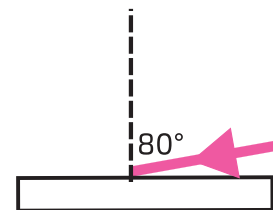
(a)



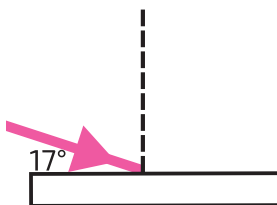
(b)



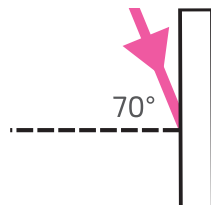
(c)



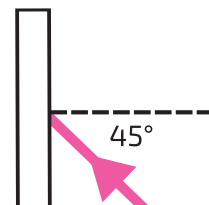
(d)



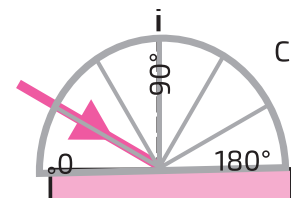
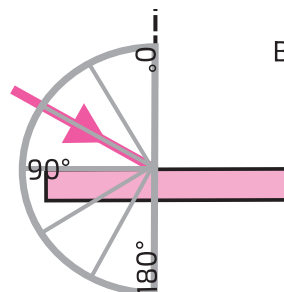
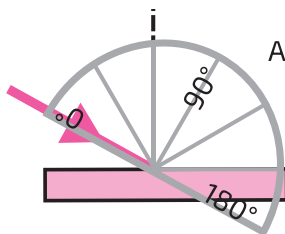
(e)



(f)

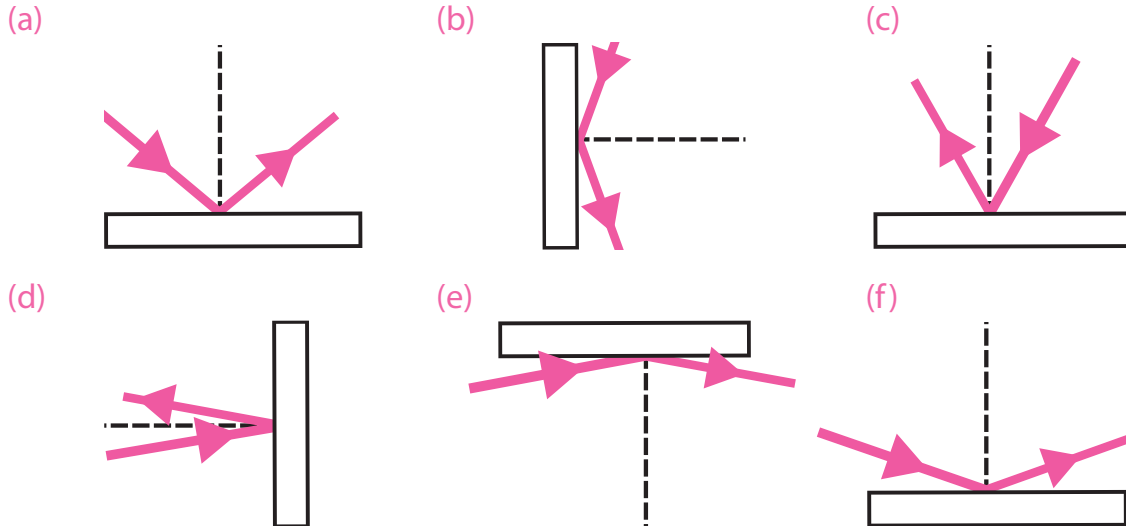


- 7 The diagrams below show three attempts to measure an angle of incidence for the same light ray.



- (a) From which of these protractors could we **not** read off the correct angle?
 (b) State the angle of incidence of the ray.
 (c) What do we line the centre of the protractor up with?

- 8 Measure the angles of incidence and the angles of reflection using a protractor.



- 9 An experiment is set up where a laser is reflected off a surface then absorbed by a receiver. Due to the camera angle, you cannot see either the laser emitter or receiver. The laser is turned on the entire time.

•**Video 1:** The original recording.

•**Video 2:** The original recording, but played in reverse.

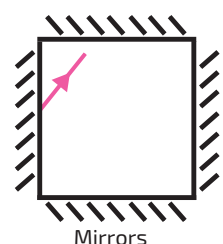
•**Video 3:** A new recording where the laser pointer and target have swapped sides.



The camera is zoomed so that you can only see the beam and the mirror — not the pointer or the target. Is it possible to see the difference between the videos? Why?

- 10 A ray is fired at two surfaces with the same angle of incidence each time. Surface A reflects all the energy, while Surface B absorbs 30 % of it and reflects the rest. The reflection is specular in both cases. What can we say about the angle of reflection for each surface?

- 11 Four large vertical mirrors are set up in a square shape. A small laser pointer is attached to one of the mirrors and pointed at another with an angle of incidence of 45° . What shapes could the laser trace as it bounces off the mirrors with this set up?



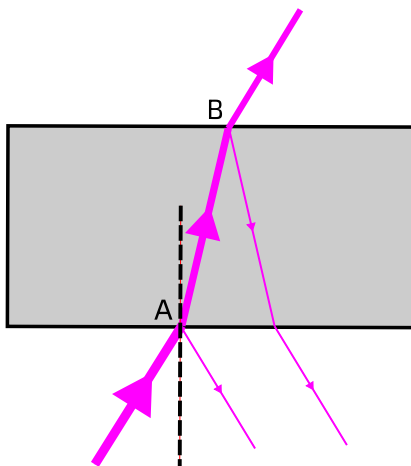
Refraction

When light travels into a **transparent** material, it changes **speed**.

This can cause it to change **direction**.

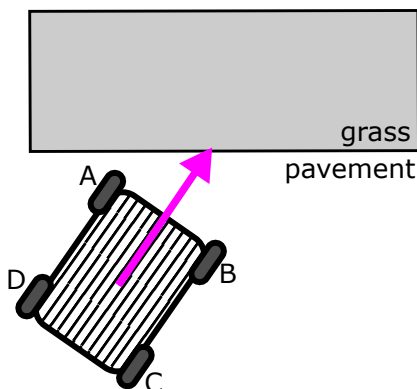
This is called **refraction**.

- 1 The diagram below shows a ray of light going through a glass block. Look at the diagram carefully, and then answer the questions.



- (a) Draw on the diagram the path the light would take in the glass if it did not change direction at A.
- (b) Label the normal with a letter N.
- (c) When the light enters the block, which way does it turn? Choose from: **towards the normal** or **away from the normal**.
- (d) Describe the direction the light takes as it moves away from the block at B.
- (e) Not all light enters the block. What happens to the light which does not enter the block?

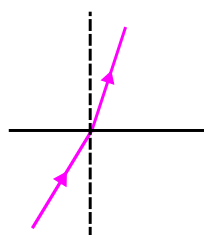
- 2 Refraction is like a trolley being pushed off a pavement onto some grass.



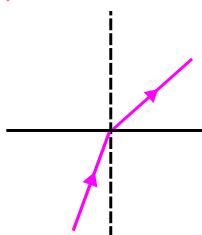
- (a) Which wheel on the trolley is going to go onto the grass first? A, B, C or D.
- (b) Which way will the trolley turn after the first wheel goes onto the grass and slows down? **Left** or **right**?
- (c) What happens after the next wheel goes onto the grass?
- (d) Draw an arrow to show the direction of the trolley after two wheels are on the grass.
- (e) Draw a normal on the diagram.
- (f) When the trolley goes onto the grass, which way does it turn? Choose from: **towards the normal** or **away from the normal**.
- (g) Compare this diagram with the diagram in question 1. Does light slow down or speed up when it enters glass from air?

- 3 A teacher drew diagrams showing light crossing a boundary from air to glass or glass to air, but forgot to label it! On each diagram, mark with a G the side which is glass.

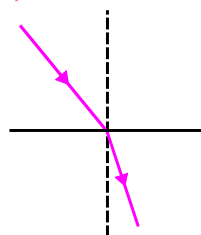
(a)



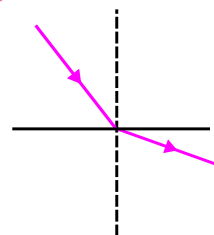
(b)



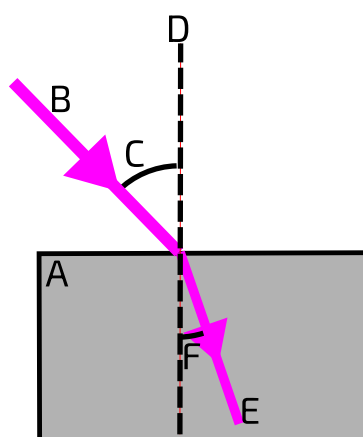
(c)



(d)

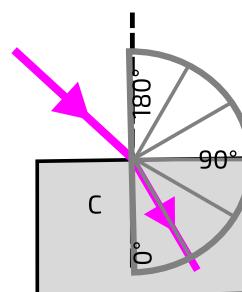
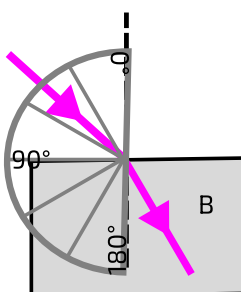
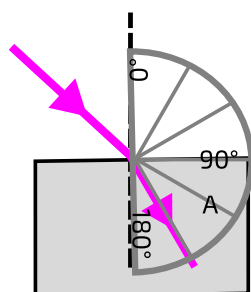


- 4 Which words describe the features in the diagram below? Fill in the table.



Technical term	A, B, C, D, E or F?
glass block	
refracted ray	
incident ray	
angle of refraction	
angle of incidence	
normal	

- 5 The diagrams below show three attempts to measure an **angle of refraction**.



- (a) From which of these protractors could we read off the correct angle?

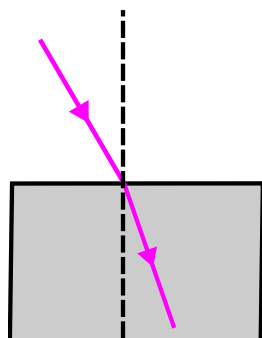
- (b) State the angle of refraction of the ray.

- (c) Estimate the angle of incidence of the ray. Is it 30° , 45° or 60° ?

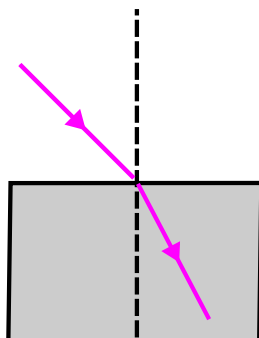
- (d) Not all of the light refracts. Draw a reflected ray onto the diagram.

- 6 In each diagram below, measure and write down the angles of incidence and refraction.

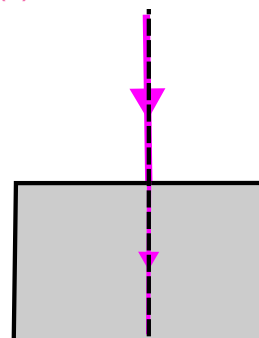
(a)



(b)

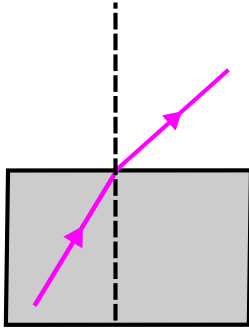


(c)

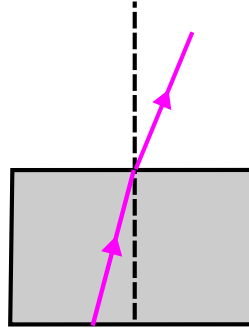


- 7 In each diagram below, write down the angles of incidence and refraction. Here the angles of incidence are inside the block of glass.

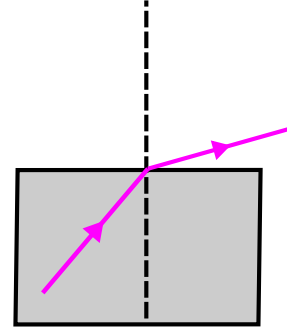
(a)



(b)



(c)



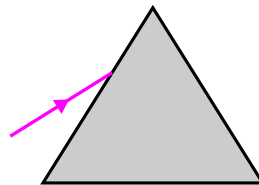
- 8 Complete the sentences using the phrases **will always**, **does not**, **in to**, **out from**, **incidence**, **refraction**. You do not need to use all of the phrases.

If the angle of incidence is zero, the light _____ change direction.

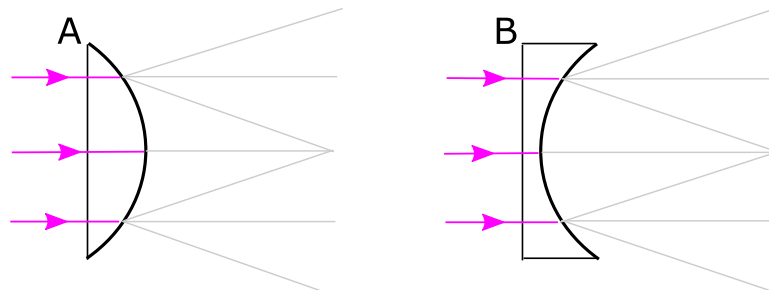
Otherwise, the angle of incidence is larger than the angle of _____ when light passes _____ a glass block.

The angle of _____ is smaller than the angle of refraction when light passes _____ a glass block.

- 9 Draw two further rays on the diagram below to show where the light might go after entering this triangular block (called a **prism**).



- 10 The diagrams below show rays passing into pieces of glass with a curved surface.



- (a) Draw normals for each of the places where light reaches the curved surface.
- (b) Draw the refracted rays to show how light leaves the glass. Each refracted ray follows one of the pale grey lines, but you will need to choose which one to use.
- (c) These objects are called **lenses**. A **convex** lens is thicker in the middle than at the edge. Which lens is the convex lens? Write A or B.
- (d) Complete the sentence: A convex lens causes parallel rays of light to

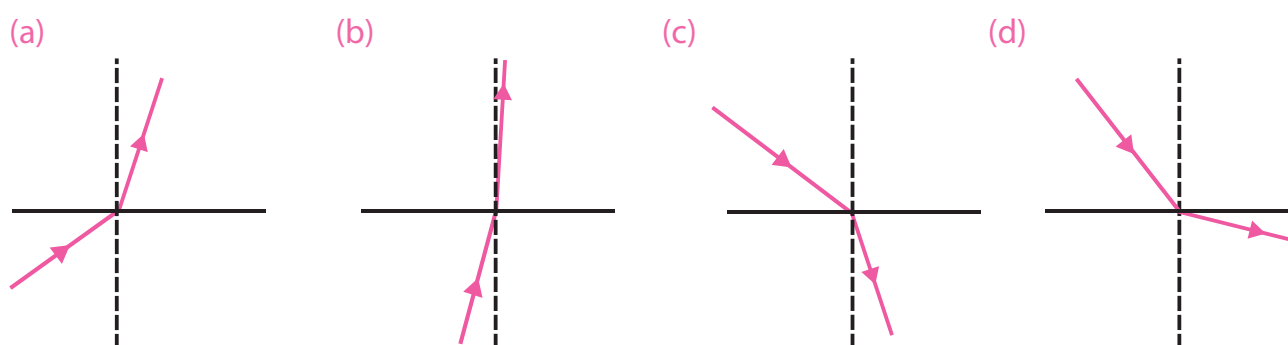
Refraction Practice

- 1 (a) When light crosses a surface from one transparent material to another, does it **always**, **sometimes**, or **never** refract? _____

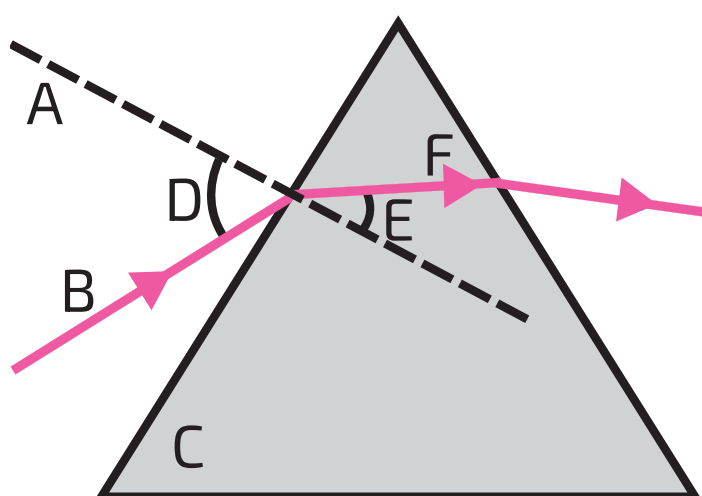
(b) Which factors decide **whether** refraction happens when light crosses a surface between materials?

- 2 A careless teacher has again forgotten to label their ray diagrams! One side of the boundary is air, one is glass.

Can you figure out which way around each of them goes? In the diagrams below, label each side of the surface.

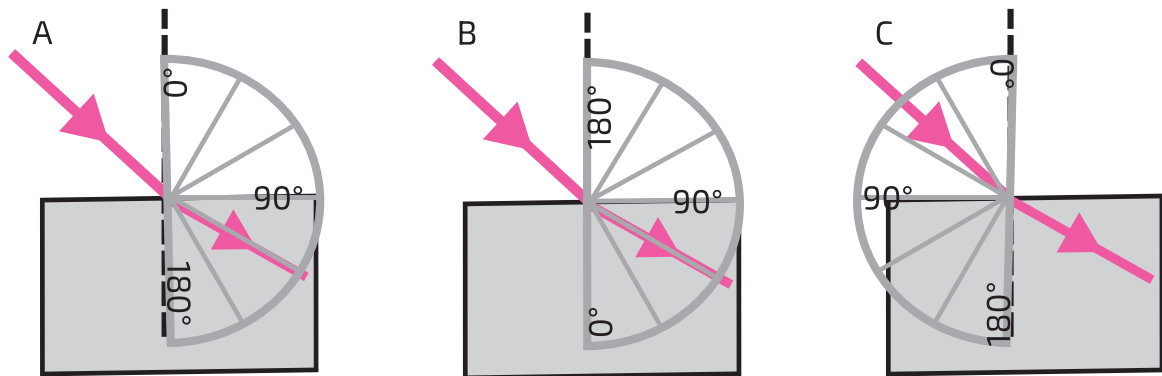


- 3 Which words describe the features in the diagram below? Fill in the table.

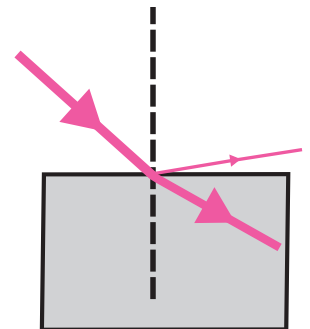


Technical term	Letter
glass block	
refracted ray	
incident ray	
angle of refraction	
angle of incidence	
normal	

- 4 The diagrams below show three attempts to measure an **angle of refraction**

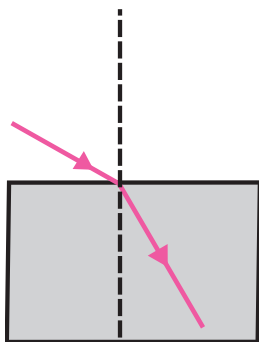


- (a) From which of these protractors could we read the correct angle?
- (b) State the angle of refraction of the ray.
- (c) Estimate the angle of incidence of the ray.
- (d) Not all the light refracts. In the diagram on the right, someone has tried to draw the reflected ray. Spot and label the mistake.

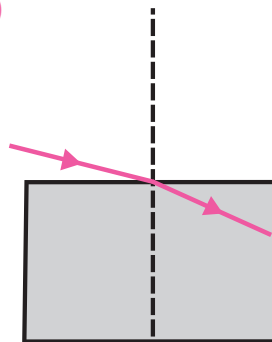


- 5 In each diagram below, write down the angles of incidence and refraction.

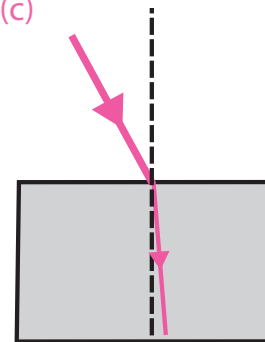
(a)



(b)

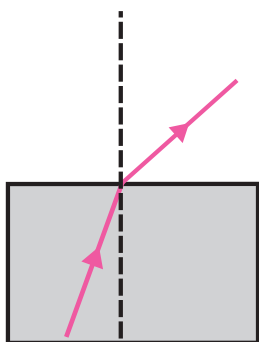


(c)

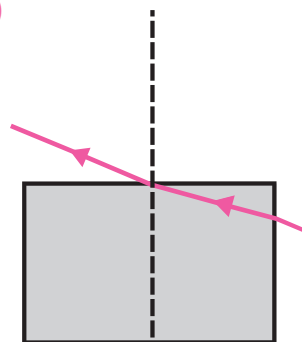


- 6 In each diagram below, write down the angles of incidence and refraction. Here the angles of incidence are inside the block of glass.

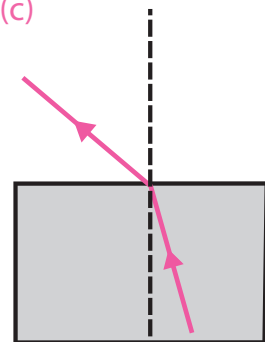
(a)



(b)



(c)



- 7 Complete the sentences using the phrases **fastest**, **slowest**, **towards**, **away from**, **speeds up**, **slows down**, and **change**. You do not need to use all of the phrases. It may help you to draw a diagram.

Light travels _____ in a vacuum and slows down when it enters glass.

If light hits glass at an angle, the side of the wave that hits first _____ first, causing the beam to _____ direction.

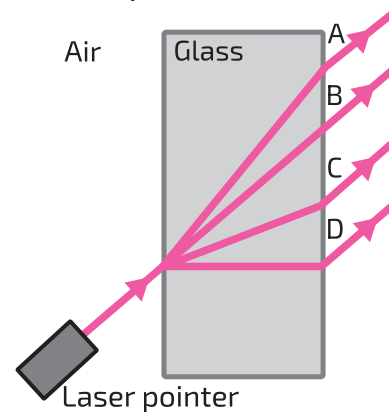
When light slows down, it bends _____ the normal.

When light speeds up as it exits the glass, it bends _____ the normal.

- 8 Some students are trying to predict the outcome of an experiment.

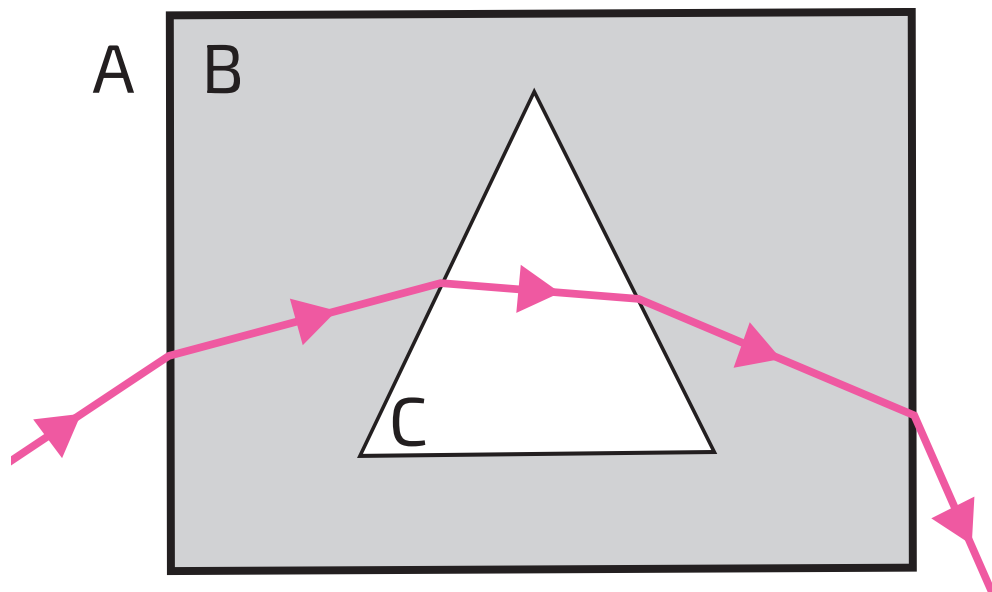
The diagram shows four different attempts at drawing what would happen when light shines through a glass block at an angle.

Which of the rays has been drawn correctly?



- 9 An antiques dealer has found a strange object in an old collection: a triangular prism sitting inside a rectangular prism.

To figure out how old it is, she wants to work out what type of materials were used to make it by studying how it interacts with light. The diagram shows a ray diagram for light passing through the object.



Order materials A, B and C according to the speed at which light travels through them, from slowest to fastest.

Waves

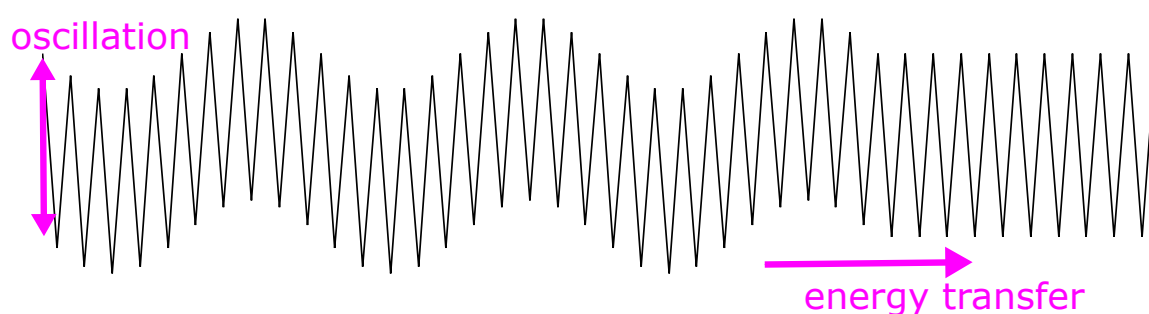
A wave carries **energy** and can carry **information** without carrying **material** from one place to another.

Inside a wave, each part moves one way then the other. This is called **oscillation**. Each part's oscillation causes the next part to oscillate as well.

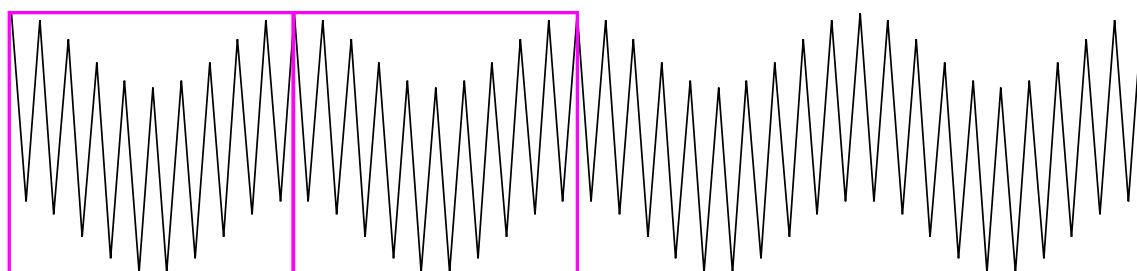
In a **longitudinal** wave the oscillations are in the **same direction** as the energy transfer.

In a **transverse** wave the oscillations are at **right angles** to the energy transfer.

- Two friends stretch a long spring between them. One of them starts moving it up and down. After a short time, it looks like this:



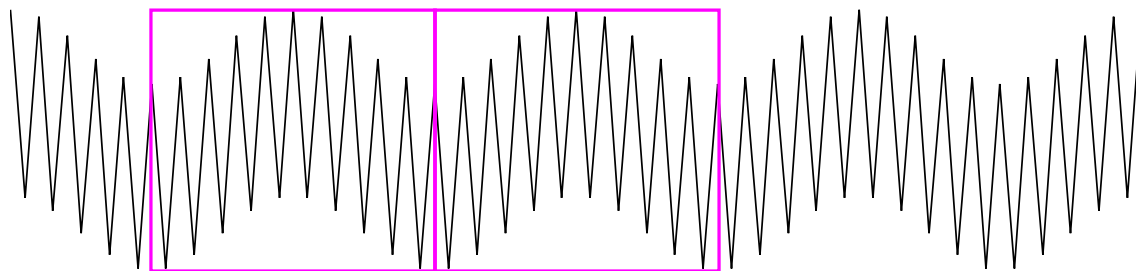
- Is this a transverse or longitudinal wave?
 - The highest points on the wave are called **peaks**. Label 3 peaks with the letter P.
 - The lowest points are called **troughs**. Label 3 troughs with the letter T.
- Use a ruler to measure the distance between neighbouring peaks in question 1.
 - Use a ruler to measure the distance between neighbouring troughs in question 1.
-
- A short time later, the wave looks like this. Boxes have been drawn around sections of the wave.



- Compare the contents of each box. Do they look the same?
- Each box contains a section of wave from one _____ to the next.
- Measure the length of each box.

(d) Draw the next box on the wave above.

5 The boxes can be drawn differently on the same wave.



(a) Each box contains a section of wave from one _____ to the next.

(b) Measure the length of each box.

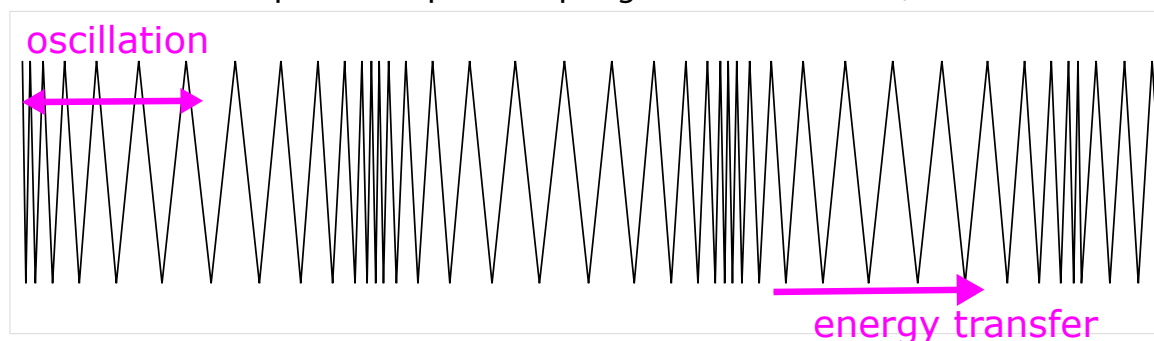
(c) Draw the next box on the wave above.

6 The boxes in questions 4 and 5 show a **repeating unit** in the wave. Their width is called the **wavelength** of the wave.

(a) Write down the wavelength of the wave.

(b) Compare your answer to (a) with your answers to questions 2 and 3.

7 The two friends now push then pull the spring. After a short time, it looks like this:



(a) Is this a transverse or longitudinal wave?

(b) The places where the spring is bunched up on the wave are called **compressions**. Label 3 compressions with the letter C.

(c) The places where the spring is most spaced out are called **rarefactions**. Label 3 rarefactions with the letter R.

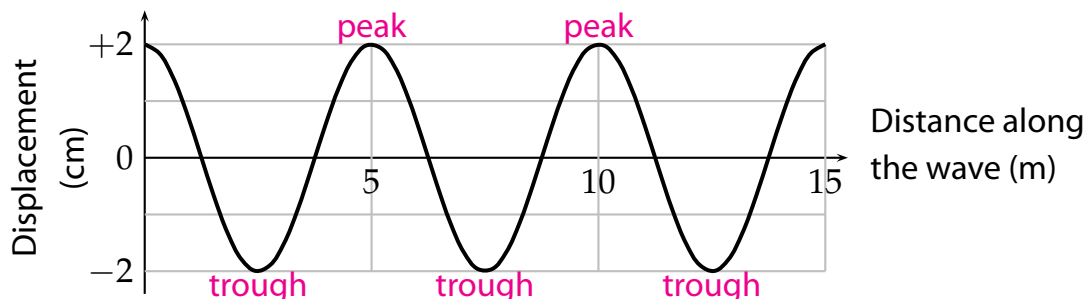
(d) Draw a box around a repeating unit on the wave.

8 Measure the wavelength of the wave in question 7.

Displacement measures how far each part of the spring is from its resting place.

When we draw a graph of a wave, we plot

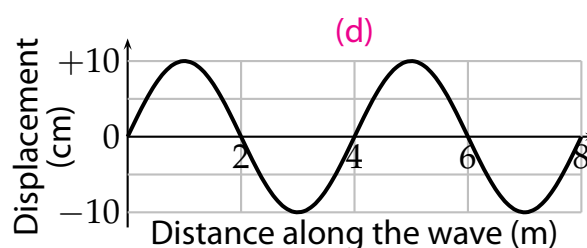
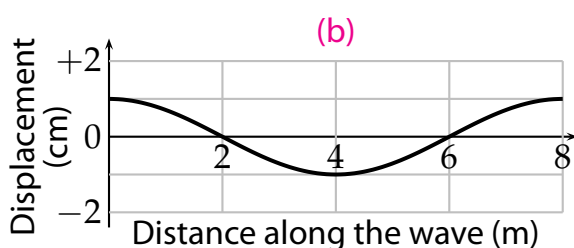
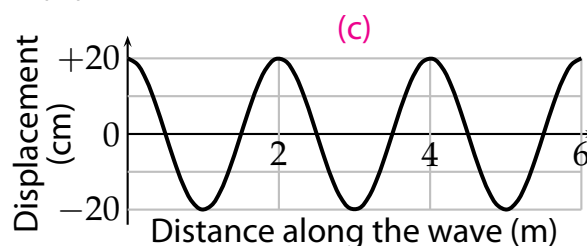
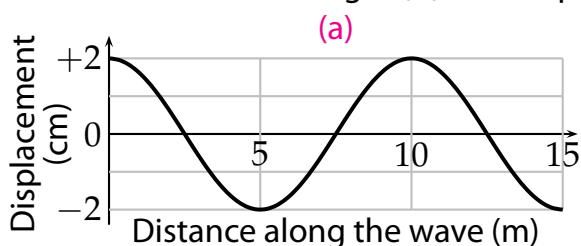
- the **displacement** on the vertical (y) axis, and
- the **distance along the wave** on the horizontal (x) axis.



The **amplitude** (A) is the largest displacement on the wave. Here, the amplitude is **2 cm**.

9 Using the scale on the graph, write down the wavelength of the wave above.

10 Write down the wavelength (λ) and amplitude (A) of each wave below.



Sound is an example of a **longitudinal** wave.

11 A citizen listens to the news on their TV. Complete the sentences using the words or phrases **towards**, **ears**, **brain**, **sound wave**, **speaker**, **oscillates**.

The _____ in the TV _____.

This makes air molecules move away from the speaker then _____ it.

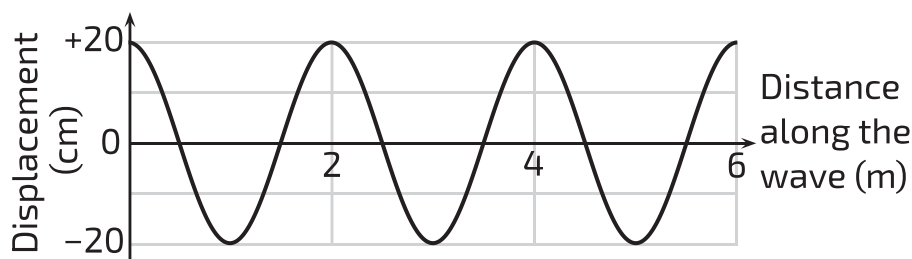
The air carries the _____ to the listener.

When the wave reaches the listener, the air makes parts of their _____ oscillate.

Nerves detect this motion, and the information reaches the listener's _____.

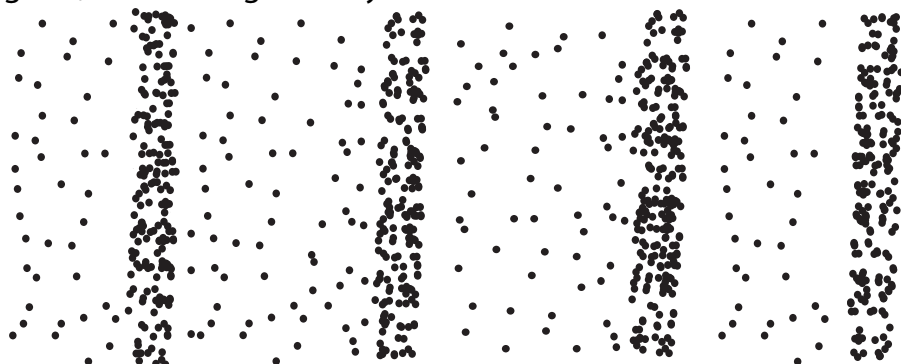
Waves Practice

- 1 Two friends stretch a long skipping rope between them. One of them starts swinging it up and down. The diagram shows what the rope looks like a short time later.



- (a) Is the wave transverse or longitudinal?
- (b) Using the letters P and T, label all the peaks and troughs on the diagram.
- 2 The diagram shows the particles in a sound wave (longitudinal) passing through a dust cloud.

On the diagram, mark a length that you would measure to measure one wavelength.



- 3 Complete the sentences using the words or phrases **L**, **oscillation**, **trough**, **peak**, **corresponding**.

To measure the wavelength (λ) of a wave, you need to measure from one point to the next _____ point. This ensures that exactly one full cycle has played out across the gap. For example, from peak to _____, or trough to _____.

If you measure horizontally from one peak to the next _____, you will only have measured half the wavelength, because only half a wave-cycle (_____) has occurred.

We use λ as the symbol for **wavelength** because it is the greek letter _____.

- 4 Complete the sentences using the words or phrases **energy**, **oscillation**, **louder**, **displacement**, **trough**, **midpoint**, **double**.

To measure the amplitude of a transverse wave, you need to measure the distance from the _____ (the equilibrium position) to a peak. This tells you how much

_____ the wave is carrying. We measure this in the direction of _____, so it looks slightly different for longitudinal and transverse waves, but is still the maximum _____ in both cases.

A common mistake is measuring all the way from a peak to a _____. Doing this will accidentally give you _____ the actual amplitude.

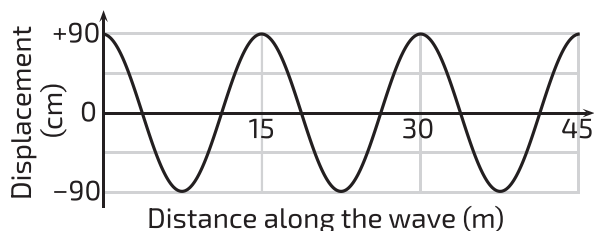
If you increase the amplitude of a sound wave, the sound will become _____, but the pitch will stay exactly the same.

- 5 Groggy the giant and Shirahoshi the giant mermaid live on a small planet called Newtonia.

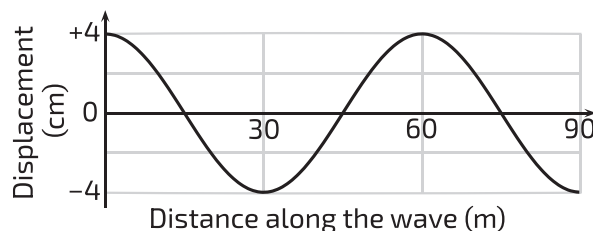
Like they do on earth, earthquakes on Newtonia create both longitudinal and transverse waves. Shirahoshi is famous for her incredibly loud voice - so much so that she can make the ground shake with her singing. Groggy says there is no way for him to detect the difference between Shirahoshi singing and a genuine earthquake.

Is he correct? Explain why.

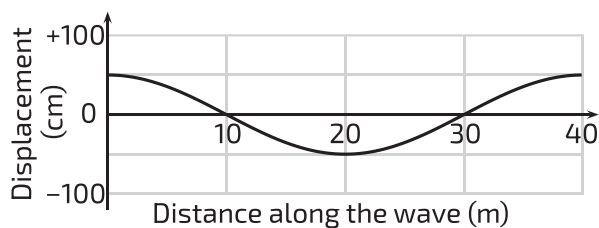
- 6 Write down the wavelength (λ) and the amplitude (A) of each wave.



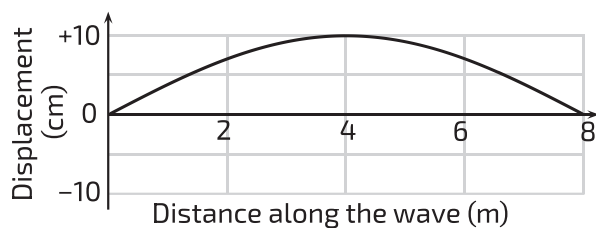
$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____



$\lambda =$ _____ $A =$ _____

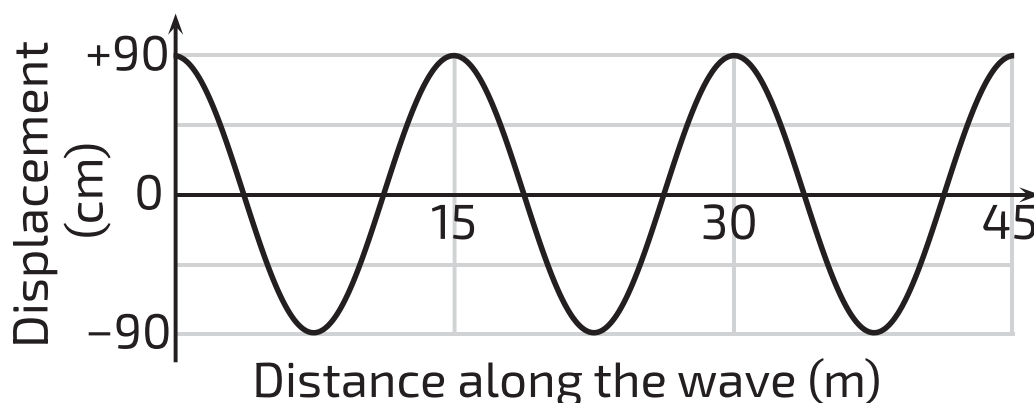
- 7 Which of these real world waves is the odd one out?

- A ripple moving across the surface of a puddle
- The sound of a teacher blowing a whistle
- A wave moving along a rope shaken up and down
- A "Mexican wave" travelling around a stadium crowd

8 Decide and label whether each of the waves below is transverse or longitudinal.

- Sound from a television
- A ripple on the surface of a lake
- A recently played guitar string
- The loud bang of a firework
- A vibrating spider web after a fly lands

9 Draw three sets of arrows showing lengths that someone could measure to measure exactly one wavelength. One arrow should be above the middle axis, one below the middle axis, and one *on* the middle axis



10 The giant mermaid Shirahoshi is singing underwater to send a message across the ocean to Groggy, who is on the shore.

To be safe, she thinks she should also write the message on paper, so a scroll is put in a bottle floating in the water directly in front of her mouth.

As she sings, her loud sound waves travel through the water and the bottle, towards the shore.

Which option(s) correctly describe the motion of the bottle?

- The bottle will stay in the same general area, bobbing back and forth in place
- The bottle vibrates because the water particles pass their energy forward without travelling far themselves
- The sound waves will push the bottle all the way to the shore where Groggy is standing
- The water particles around the bottle travel to the shore, leaving the bottle behind in a vacuum

Sound and Waveforms

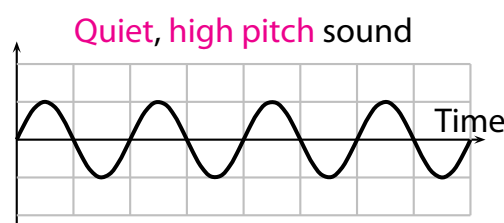
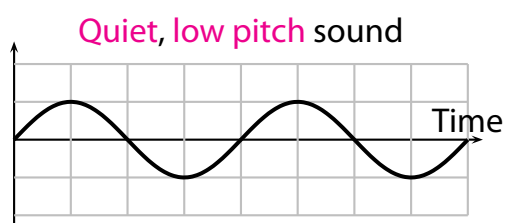
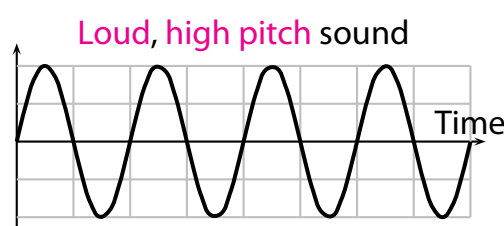
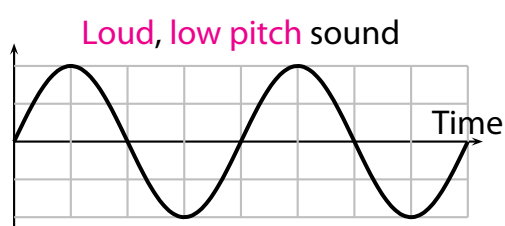
When a sound travels, the **particles oscillate** in the material it is travelling through. This means that they move forwards and backwards again and again.

The time for each oscillation back and forth is called the **time period**.

When the sound reaches our **ears** or a **microphone**, it makes a sensitive surface **oscillate** too. This can be heard as sound.

When a **microphone** is connected to a computer, a **data-logger** or an **oscilloscope**, the wave can be seen on a screen. The screen shows the **displacement** changing as **time** passes.

The greatest displacement of a wave is called its **amplitude**.



1 Look at the waves shown above and answer the questions.

(a) Describe how the loud sounds' graphs differ from the quiet sounds' graphs.

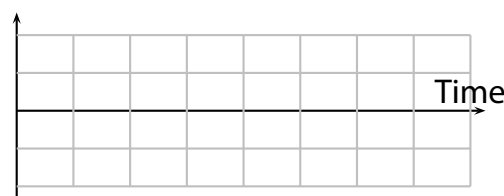
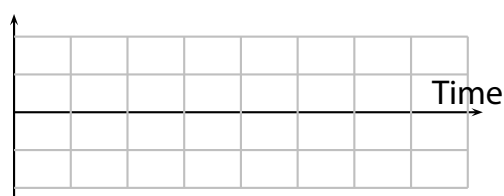
(b) Describe how the high sounds' graphs differ from the low sounds' graphs.

(c) Does a short time period mean that the sound is high pitched or low pitched?

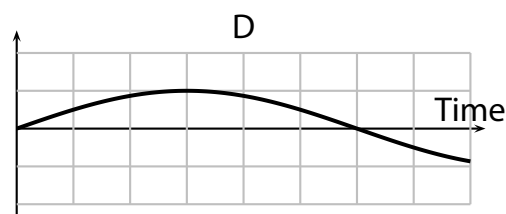
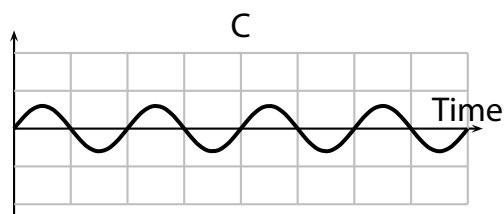
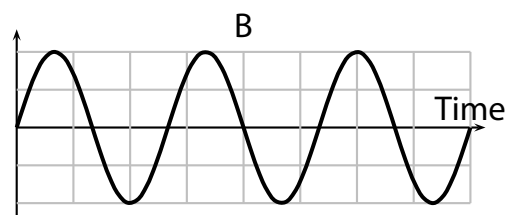
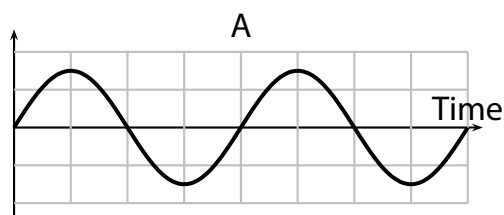
(d) Does a large amplitude mean that the sound is loud or quiet?

2 Draw waves on the graphs below to fit the descriptions.

(a) quieter and higher than the waves above (b) louder and lower than the waves above



3 Use these wave graphs to answer the questions below them.



- (a) Put the letters ABCD in order to rank the waves from quietest to loudest.
- (b) Put the letters ABCD in order to rank the waves from lowest to highest pitch.
- (c) Which wave has the shortest time period?

The number of complete oscillations in one second is called the **frequency** of the wave.

Waves with a high frequency have a **short** time period and are **high pitched**.

4 Which wave in question 3 has the lowest frequency?

5 Complete the sentences using **20 000 Hz**, **ultrasound**, **Hz**, **20 Hz**, **hertz**.

Frequencies are measured in _____ (symbol _____).

Our ears can hear sounds with frequencies between _____ and _____.

Sounds above this range can not be heard and are called _____.

Each oscillation takes a very short time so the times are measured in **milliseconds (ms)**.

A millisecond is one **thousandth** of a second. There are **1000** milliseconds in one second.

We write $1\text{ s} = 1000\text{ ms}$, or $1\text{ ms} = 0.001\text{ s}$.

6 Write the times below in milliseconds.

(a) 0.002 s

(b) 0.125 s

(c) 0.02 s

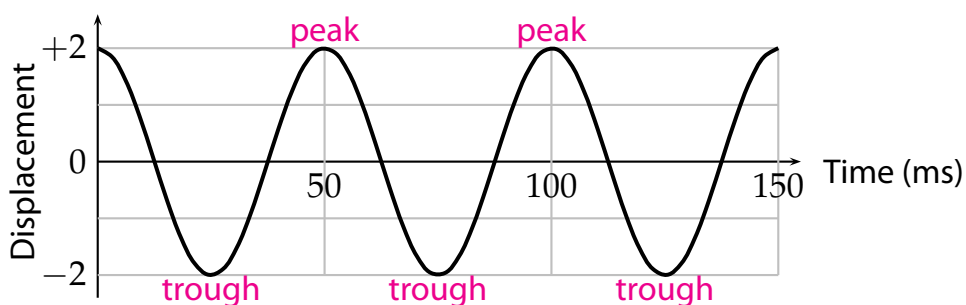
7 Write the times below in seconds.

(a) 2000 ms

(b) 3 ms

(c) 40 ms

The **time period** of this wave is 50 ms. This is the time from one **peak** to the next **peak**.



8 What is the time period of this wave in seconds?

9 In this question we look at the link between time period and frequency.

(a) A wave has a time period of 0.05 s. Work out how many times it oscillates each second using an equation.

$$\boxed{1} \div \boxed{0.05} = \boxed{}$$

(b) What is the frequency of a wave if its time period is 0.05 s?

(c) A wave has a time period of 0.002 s. Work out its frequency using an equation.

$$\begin{array}{ccccc} 1 & \div & \text{time period (s)} & = & \text{frequency (Hz)} \\ \boxed{1} & \div & \boxed{0.002} & = & \boxed{} \end{array}$$

(d) Work out the frequency of a wave with a time period of 0.004 s using an equation.

$$\begin{array}{ccccc} 1 & \div & \text{time period (s)} & = & \text{frequency (Hz)} \\ \boxed{1} & \div & \boxed{0.004} & = & \boxed{} \end{array}$$

10 A wave has a time period of 5 ms. Calculate its frequency.

11 What is the frequency of the wave in question 8?

A frequency of 4000 Hz is usually written **4 kHz**. One kilohertz (1 kHz) is the same as **1000 Hz**.

12 Calculate the frequencies in kHz of waves with the following time periods.

(a) 10 ms

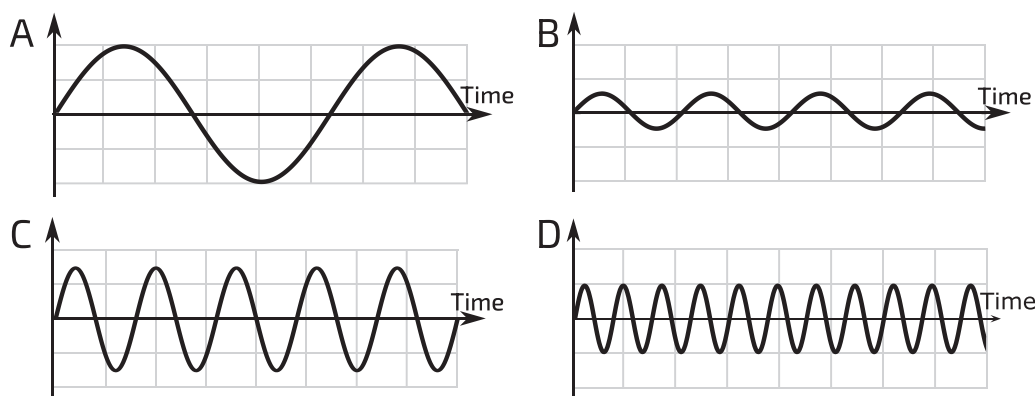
(b) 0.5 ms

(c) 40 ms

(d) 0.1 ms

Sound and Waveforms Practice

- 1 Look at the waves shown below and answer the questions.

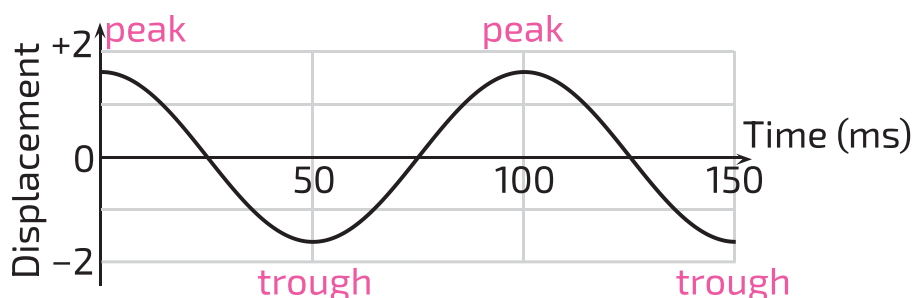


- (a) Rank the waves from loudest to quietest: _____
 (b) Rank the waves from highest to lowest pitch: _____
 (c) Which wave in Figure 1 has the shortest time period? _____
- 2 Which of the waves from question 1 has the greatest frequency? _____
- 3 Complete the sentences using **Hz, kHz, 3, 40 and 1000**. You may use each more than once.

If a frequency is very high, we might measure it in _____, which are each equal to _____.

For example, a beeping smoke alarm has a frequency of roughly _____, while an open E note on a bass guitar is roughly _____.

- 4 Write the following times in milliseconds.
 (a) 0.007 s (b) 0.301 s (c) 0.05 s
- 5 Write the following times in seconds.
 (a) 5000 ms (b) 9 ms (c) 40.1 ms
- 6 What is the time period, in seconds, of this wave?



- 7 (a) A wave has a time period of 0.2 s. Work out how many times it oscillates each second using an equation.

$$\boxed{1} \div \boxed{} = \boxed{}$$

- (b) What is the frequency of a wave if its time period is 0.2 s?

- (c) A wave has a time period of 0.008 s. Work out its frequency using an equation.

$$\frac{1}{\boxed{1}} \div \text{time period (s)} = \text{frequency (Hz)}$$

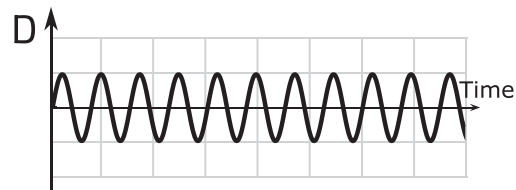
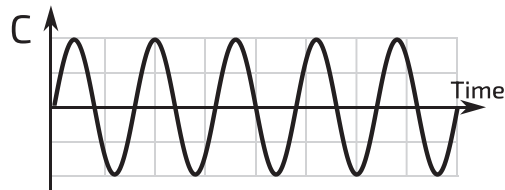
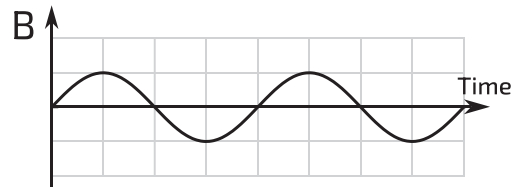
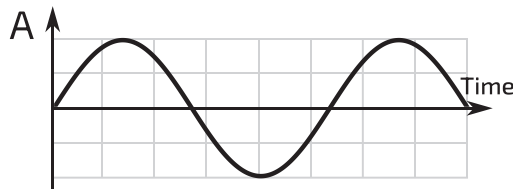
$$\boxed{1} \div \boxed{} = \boxed{}$$

- (d) Work out the frequency of a wave with a time period of 0.08 s using an equation.

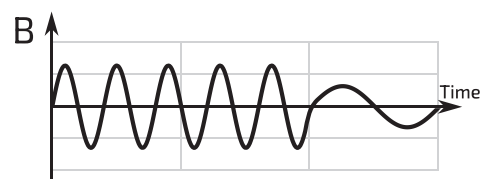
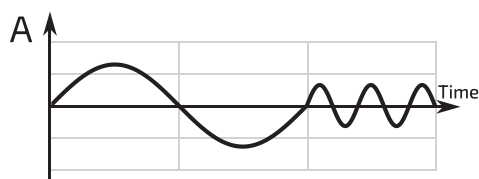
$$\frac{1}{\boxed{1}} \div \text{time period (s)} = \text{frequency (Hz)}$$

$$\boxed{1} \div \boxed{} = \boxed{}$$

- 8 A wave has a time period of 800 ms. Calculate its frequency.
- 9 What is the frequency of the wave in question 6?
- 10 Calculate the frequencies in kHz of waves with the following time periods.
- (a) 25 ms (b) 200 ms (c) 0.4 ms (d) 0.001 ms
- 11 Each graph represents the sound wave for one of the situations in the table below. Match the situation to the graph.



- (a) The buzzing from a mosquito's wings (c) The booming of thunder after lightning
- (b) A singer hitting a high note at a festival (d) A quiet bass guitar backing track
- 12 A guitarist is tuning her guitar. She plays a note and then quickly tightens the string, increasing its pitch.



Which of the graphs in Figure 12 most accurately represents the sound that the guitar makes?